

CW Practice MP3 Generator

Build Your Own Morse Code Ear Training Audio — Morse Ninja Style

If you've used the Morse Ninja YouTube videos to practice CW, you know the format: a word is sent in Morse code three times, then a voice announces the word, then the code plays one final time. It's a proven method for building head copy skills.

This is a free, open-source Python script that generates those same style practice MP3 files on your own computer. Every time you run it, the word list is shuffled into a new random order so you never memorize the sequence. You can adjust the speed, tone pitch, and Farnsworth spacing to match your skill level.

The word list is the top 150 most common English words (from sightwords.com), which gives you a solid foundation for recognizing everyday language at speed. You can also swap in your own word list — Q-signals, callsign prefixes, contest exchanges, whatever you want to drill.

What It Sounds Like

Each word follows this pattern:

CW CW CW → [voice says the word] → CW → [beep] → [silence] → next word

A short 1000 Hz beep marks the end of each word group, and a comfortable pause follows before the next word begins. The full 150-word file runs about 30-40 minutes depending on your speed settings — perfect for a hike, a drive, or a session at the bench.

What You Need

You need three things installed on your Windows PC. All are free.

Python — the programming language that runs the script. Download from python.org. During installation, check the box that says "Add Python to PATH" — this is important.

ffmpeg — converts the audio to MP3 format. Open a Command Prompt or PowerShell window and type:

```
winget install ffmpeg
```

Two Python packages — numpy (for generating the CW tones) and pytsx3 (for the spoken voice). Open a Command Prompt and type:

```
pip install numpy pytsx3
```

That's it for setup.

Installation

1. Create a folder somewhere convenient, like C:\Users\YourName\cwbot
2. Download or copy these two files into that folder:
 - **cwgen.py** — the generator script
 - **words.txt** — the 150-word list (one word per line)
3. The words.txt file contains the top 150 English words from sightwords.com. I found this list at https://sightwords.com/pdfs/word_lists/top_150_all.pdf — just use the provided words.txt file as-is.

Creating Your First MP3

Open a Command Prompt, navigate to your folder, and run:

```
python cwgen.py
```

This takes a few minutes to generate all 150 words with spoken announcements. When it finishes, you'll see the file in the queue folder ready to copy to your phone or music player.

Command Line Options

```
python cwgen.py [OPTIONS]
```

--wpm N	Character speed in WPM (default: 30)
--farn N	Farnsworth overall speed in WPM (default: 23)
--count N	Use the first N words from the list (default: all 150)
--freq N	CW tone frequency in Hz (default: 700)
--voice-speed N	Voice speaking rate, lower is slower (default: 130)
--no-voice	CW tones only, skip spoken words (good for testing)

Examples

Generate at a comfortable learning speed:

```
python cwgen.py --wpm 20 --farn 15
```

Challenge yourself at higher speed:

```
python cwgen.py --wpm 35 --farn 28
```

Lower pitch sidetone (easier on some ears):

```
python cwgen.py --freq 600
```

Just the first 50 words for a shorter session:

```
python cwgen.py --count 50
```

Output Files

MP3 files are saved in the queue subfolder with descriptive names:

```
queue/cw_30wpm_23f_tone700.mp3  
queue/cw_25wpm_20f_tone600.mp3
```

The filename shows the character WPM, Farnsworth WPM, and tone frequency so you can tell your files apart at a glance. Running the same settings again overwrites the previous file (the word order will be different since it's reshuffled each time).

Understanding Farnsworth Spacing

The script uses Farnsworth timing, which is how most CW training tools work. The individual characters are sent at full speed (the --wpm setting), but the gaps between characters and between words are stretched out to give you more time to process what you heard. The --farn setting controls the overall effective speed.

For example, --wpm 30 --farn 23 sends each letter at 30 WPM but spaces them so the overall throughput is 23 WPM. As your skills improve, bring the Farnsworth speed closer to the character speed until they match — then you're copying at full speed.

Customizing the Word List

The words.txt file is plain text, one word per line. You can edit it to focus on whatever you want to practice. Some ideas:

- Common Q-signals: QTH, QSL, QRZ, QSO, QRM, QRN, QSB, QRP, QRO
- Contest exchanges: five nine, five nine nine, five ten
- Greeting phrases: hello, good morning, good afternoon, good evening
- Callsign prefixes: whiskey, alpha, kilo, november
- Your weak spots — words you consistently miss

Save the file and run the script again. It will use whatever words it finds in words.txt.

Troubleshooting

"python is not recognized" — Python wasn't added to your PATH during installation. Reinstall Python from python.org and make sure to check "Add Python to PATH."

"ffmpeg is not recognized" — ffmpeg isn't installed or isn't in your PATH. Try "winget install ffmpeg" and restart your Command Prompt. Alternatively, download ffmpeg from ffmpeg.org, extract it, and add the bin folder to your system PATH.

"No module named numpy" or "No module named pytsx3" — Run "pip install numpy pytsx3" from a Command Prompt.

Voice sounds robotic — The script uses your Windows built-in text-to-speech voice. Windows 11 typically has "Microsoft David" (male) and "Microsoft Zira" (female). The quality is decent for single-word announcements. You can install additional voices through Windows Settings > Time &

Language > Speech.

File takes a long time to generate — The full 150-word file with voice takes 2-5 minutes depending on your PC. Use --no-voice for faster generation if you just want CW tones, or --count 50 for a shorter file.

Credits

Word list: Top 150 Written Words from sightwords.com

https://sightwords.com/pdfs/word_lists/top_150_all.pdf

Training format inspired by Morse Ninja (YouTube)

<https://www.youtube.com/@MorseNinja>

Files can be downloaded from Kevin's Dropbox here:

<https://www.dropbox.com/scl/fo/3by74qnjv70c56plba9p9/AIUBRODScQREFDa2IUTTLho?rlkey=ii9b03u5smtyygfm1v316o0i1&dl=0>

or short URL: <https://tinyurl.com/CW-Gen-AJ6EE>

Script by Kevin AJ6EE — built with Claude AI

73 de AJ6EE